

Abstract

The Free Energy Principle (FEP) and Active Inference have expanded rapidly across neuroscience, robotics, biology, and formal mathematics—supported by new theoretical foundations in Bayesian mechanics and path-integral formulations. However, the field faces a dual challenge: the sheer volume of publications makes systematic synthesis difficult, while the FEP's theoretical generality has invited falsifiability critiques, since its core claim can be applied to any self-organizing system without clearly distinguishing itself from alternatives. Both challenges point to the same methodological gap: a lack of systematic, hypothesis-specific evidence profiling across the field's core claims. Building on the systematic literature analysis of Knight, Cordes, and Friedman knight2022fep—which pioneered manual annotation paired with ontology-based analysis at the scale of hundreds of papers—we present a computational meta-analysis framework that automates and scales this approach. Our pipeline retrieves literature from arXiv, Semantic Scholar, and OpenAlex, deduplicating records via a canonical identifier hierarchy. It classifies papers into a three-tier